SOP-IMG-014 Immunofluorescence staining

Fictional laboratory protocol · version 3.2 · for software testing

1 Purpose and scope

Standard procedure for immunofluorescence staining of fixed brain sections (fictional). Applies to cryosections of 20-40 um thickness. Keep slides protected from light after step 6.

2 Reagents

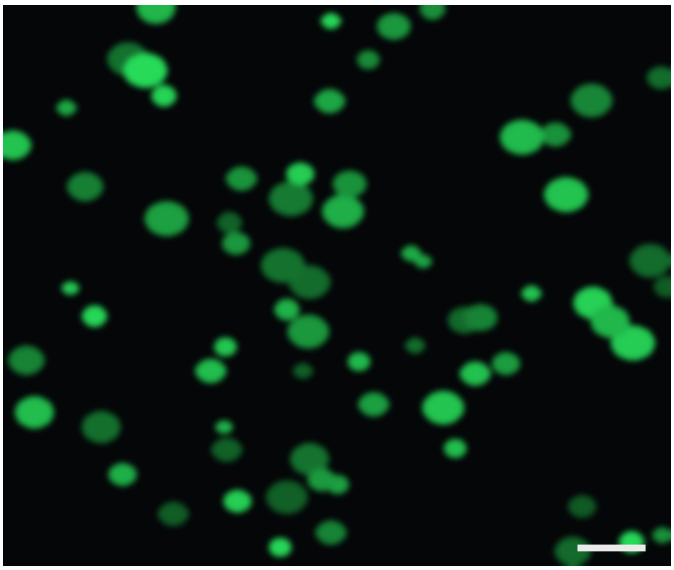
Reagent	Supplier (fictional)	Catalogue no.	Working dilution
Anti-Iba1 (rabbit)	Example Bio	EB-0019	1:500
Anti-GFAP (chicken)	Sample Reagents	SR-4410	1:1000
Goat anti-rabbit AF488	Example Bio	EB-7731	1:800
DAPI	Sample Reagents	SR-0002	1 ug/mL
Normal goat serum	Example Bio	EB-1200	5 % (v/v)

3 Procedure

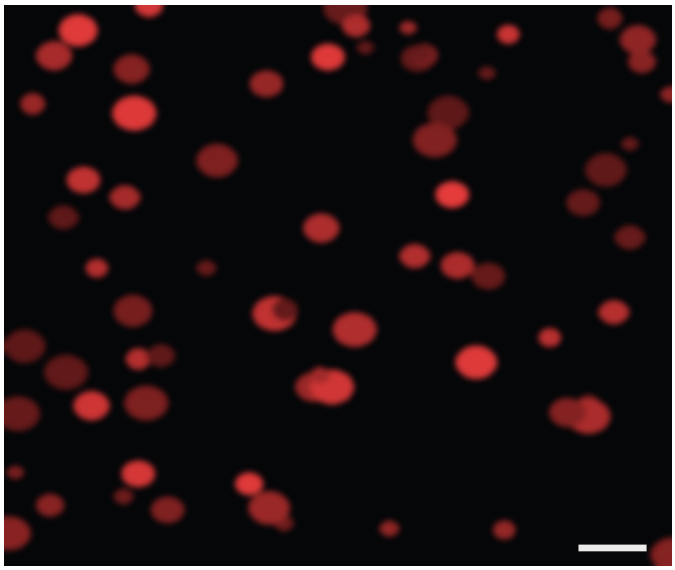
1. Wash sections 3 x 5 min in PBS (pH 7.4) at room temperature.
2. Permeabilise in 0.3 % Triton X-100 in PBS for 15 min.
3. Block in 5 % normal goat serum for 1 h at 20-25 C.
4. Incubate with primary antibodies overnight at 4 C.
5. Wash 3 x 10 min in PBS.
6. Incubate with secondary antibodies for 2 h, protected from light.
7. Counterstain with DAPI for 10 min; wash 2 x 5 min.
8. Mount with anti-fade medium; store at 4 C in the dark.

CAUTION Paraformaldehyde is toxic. Work in a fume hood and wear gloves. This warning is part of a fictional test document.

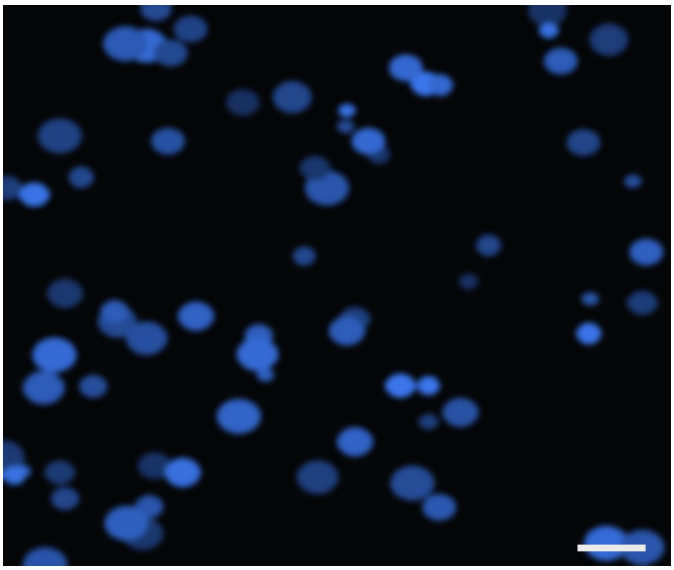
4 Representative images



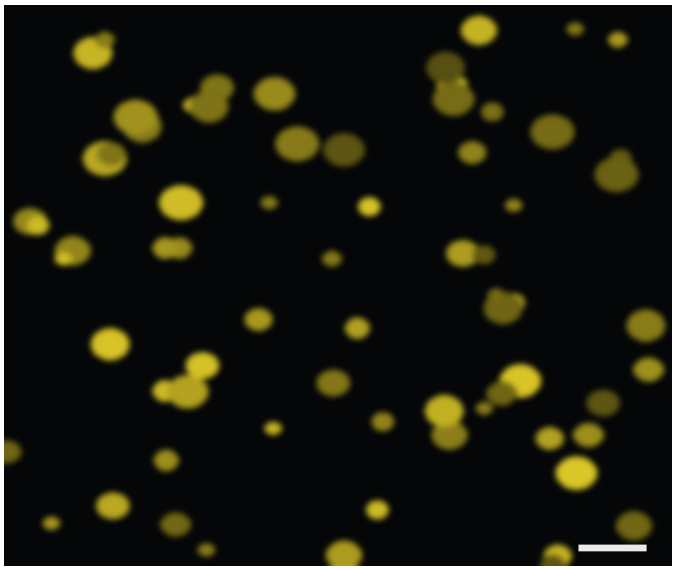
A Iba1 (green)



B GFAP (red)



C DAPI (blue)



D Merge

Figure 1. Fictional confocal images of mouse hippocampus (CA1), 20x objective. Scale bar 50 um. Images are synthetic and generated for software testing only.

5 Differential expression (example)

Aged vs young microglia · fictional values · FDR < 0.05

Gene	log2 fold change	baseMean	p-value	padj	Direction
Apoe	+2.41	5,120.4	1.2e-12	4.8e-11	up
Cst7	+3.05	812.7	4.5e-10	1.8e-08	up
Lgals3	+1.88	1,490.2	2.1e-08	8.4e-07	up
Itgax	+2.67	233.9	7.7e-08	3.1e-06	up
P2ry12	-1.42	9,830.1	3.3e-07	1.3e-05	down
Tmem119	-1.15	7,210.6	1.9e-06	7.6e-05	down
Cx3cr1	-0.92	6,650.0	8.4e-05	3.4e-03	down
Spp1	+2.12	418.3	2.6e-04	1.0e-02	up

